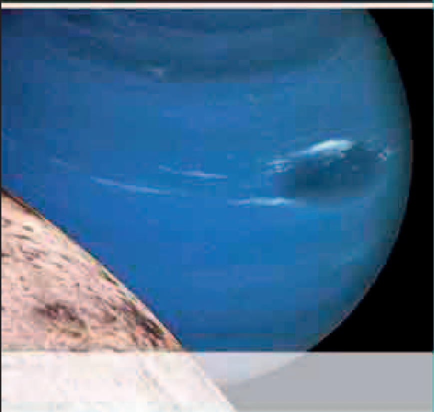
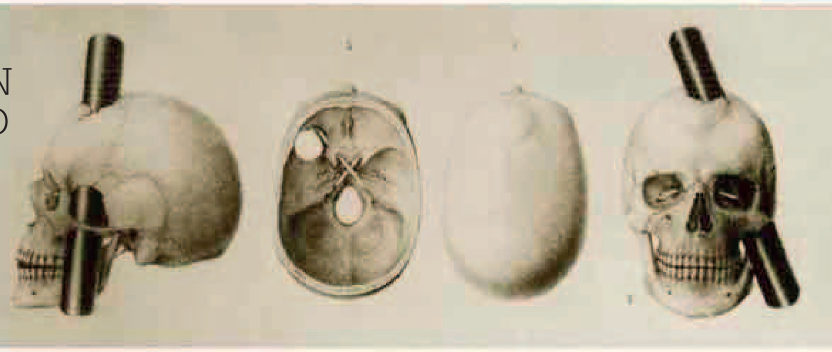


> **“** THE **EQUILIBRIUM…** BETWEEN HIS INTELLECTUAL FACULTIES AND ANIMAL PROPENSITIES… [HAS] BEEN **DESTROYED. ”**
>
> **John Martyn Harlow, American physician,** in *Recovery from the Passage of an Iron Bar through the Head,* 1868



Following an iron bar penetrating Phineas Gage’s skull and brain, studies on how his personality changed after this accident revealed much about brain function.

Advances in anesthesia were made by surgeons on both sides of the Atlantic. On October 16, at the Massachusetts General Hospital in Boston, American surgeon **William Morton** (1819–68) anesthetized his patient, Gilbert Young, **sending him to sleep with fumes from ether** while Morton cut out a tumor from his neck. Half an hour later Young woke up, unaware the operation had been done. Others had used anesthetics before, such as American surgeon **Crawford Long** (see 1842–43) and American dentist **Horace Wells**, but it was Morton’s demonstration that made an impact. Two months later, Scottish surgeon **Robert Liston** (1794–1847) performed a leg amputation under anesthetic in London.

Another important discovery was the production of **kerosene**, or paraffin, **from coal or oil**, by Canadian geologist **Abraham Pineo Gesnar** (1797–1864). In 1846, Gesnar began experimenting with methods for distilling coal and oil. By 1853, he had perfected a process to produce a new fuel he named kerosene that was used in lamps. Up until this time most lamps were fuelled by whale oil, but kerosene was much cheaper, so people could afford to burn lights brighter and longer.

IN 1847 Hungarian physician **Ignaz Semmelweis**. made an important discovery for medicine. He realized that if **doctors washed their hands** this could **reduce infection** by puerperal fever: a disease responsible for the deaths of many woman during childbirth. His procedures were not adapted until many years later.

Scottish surgeon **James Simpson** realized that neither ether nor laughing gas could keep a patient unconscious long enough for a long operation, and introduced **chloroform** as an anesthetic.

German physicist **Hermann von Helmholtz** outlined the **law of the conservation of energy**, which was first put forward by **Julius von Mayer** (see 1841). This stated that energy cannot be created or destroyed (now known as the first law of thermodynamics).

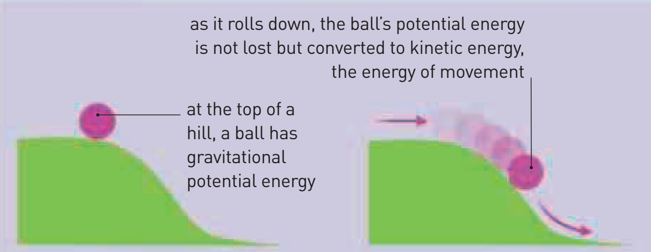
The following year, Scottish physicist **William Thomson** (Lord Kelvin) formulated the third law of thermodynamics with his **idea of absolute zero**. He realized there must be a temperature at which all molecular movement ceases and calculated it to be −459.67°F (−273.15°C). Thomson used this as the starting point for a new temperature scale—the **Kelvin scale** (see 1740–42).

By 1848, increasingly powerful telescopes were revealing more about the solar system. Astronomers discovered **planets** often had **more than one moon**. **William Lassell** and American astronomer **William Bond** (1789–1859) discovered Saturn’s eighth moon, Hyperion.

English inventors **John Stringfellow** (1799–1883) and **William Henson** (1812–88) flew a model of their steam-powered aircraft, the Aerial Steam Carriage, for 33ft (10m). It was the **first ever powered flight**, but attempts to fly a larger model were unsuccessful.

Vermont railroad worker Phineas Gage survived a 3.3ft (1m) iron rod being driven through his head, resulting in intellectual and personality changes. This was the first record of how damage to the frontal lobe of the brain affected function.

Chloroform inhaler
The chloroform inhaler was developed in 1848. It enabled surgeons to quickly anesthetize patients with fumes of vaporized chloroform.



CONSERVATION OF ENERGY

The law of the conservation of energy states that the total amount of energy in the universe is constant. Energy cannot be created nor destroyed, it can only change form. When any work occurs, energy is converted from one form to another or transferred from one object to another. For example, potential energy (in a static object) can be converted to kinetic energy (motion energy).

October 16 American surgeon William Morton performs an operation under **general anesthesia**

December 21 Scottish surgeon Robert Liston performs a leg amputation under **general anesthetic**

1847 German physicist Hermann von Helmholtz formally states the **law of conservation of energy**

1847 Hungarian doctor Ignaz Semmelweis recommends **washing hands to reduce childbirth infections**

October 1, 1847 American astronomer Maria Mitchell discovers **comet 1847 VI**

November 4–8, 1847 Scottish surgeon James Simpson uses **chloroform** for an **operation**

1848 Scottish physicist William Thomson calculates the **absolute zero point of temperature**

June 1848 English inventors John Stringfellow and William Henson make the **first powered flight**

September 13, 1848 American railroad worker **Phineas Gage** survives an iron rod penetrating his head

September 20, 1848 The American Association **for the Advancement of Science** is established